

Database Project

Database Management Systems Course Project

Spring Semester – Academic Year 2025/2026

Dr. Belal Alfara

Eng. Ahmed Alkahlout

Design and Implementation of a Real-World Database Management System

Project Description

Students are required to design and implement a complete database system for a real-world application. Examples include (but are not limited to):

- University Management System
- Hospital Management System
- Library System
- E-commerce System
- Banking System

The project must follow a structured database engineering approach, moving from problem analysis to a fully functional SQL implementation.

Technical Requirements

1. Problem Description

- Clear description of the system
 - Identify:
 - Users
 - Functional requirements
 - Objectives
-

2. ER Diagram (Conceptual Design)

- Must include:
 - **5–7 entities minimum**
 - **At least one M:N relationship**

3. Relational Schema (Logical Design)

- Each table must include:
 - Primary Key
 - Foreign Keys

4. Normalization

- Show full process:
 - $UNF \rightarrow 1NF \rightarrow 2NF \rightarrow 3NF$
- Use real examples from your design

5. SQL Implementation (Physical Design)

Required:

- DDL:
 - CREATE TABLE
- DML:
 - INSERT
- Queries:
 - 5 queries only

6. Results

- Must include 4 screenshots of:
 - Table creation
 - Data insertion
 - Query execution
 - Output results
-

Naming Convention (Required)

1. Use **singular names for entities** (Student, Course, ...)
2. Use **lowercase and underscores for SQL tables**
(student, course_registration)
3. **Primary keys format:** entity_id
4. **Foreign keys must match referenced primary key**
5. Use **clear attribute names** (first_name , but not fname )
6. **Consistency across ERD, schema, and SQL is mandatory**

Required Submissions (What Students Must Submit)

Students must submit **all of the following files**:

1. Project Report (Word Document)

A single organized document that includes:

- Title Page (Project title, student name, ID)
 - Introduction
 - Problem Description
 - ER Diagram (inserted as image)
 - Relational Schema (tables clearly written)
 - Normalization (step-by-step with examples)
 - SQL Implementation (code sections)
 - Results (screenshots with captions)
-

2. ER Diagram File

- Submitted separately as:
 - Image (PNG/JPG) **or**
 - PDF

3. SQL Scripts (Separated Files)

Students must submit **SQL files (1 – 2 files only)**, each with a clear purpose:

4. PowerPoint Presentation

Must include:

- Project overview
 - Problem description
 - ER Diagram
 - Schema
 - Normalization summary
 - SQL highlights
 - Results
-

5. Final Presentation

- Duration: **10-15 minutes**
 - Requirements:
 - Explain the system clearly
 - Justify design decisions
 - (Optional) Live SQL demo
 - All students must participate
 - Includes Q&A session
-

Grading Rubric

Component	Marks
Problem Description	10
ER Diagram	20
Relational Schema	15
Normalization	15
SQL Implementation	25
Results	5
Presentation	10
Total	100

Work Policy

- Individual

Academic Integrity

- Copying is strictly prohibited
- Submissions will be checked for similarity
- Violations will be penalized

Additional Notes

- The project should be **include 5-8 tables**
- Avoid overly simple designs
- Copying is strictly prohibited
- Submissions that rely heavily on AI-generated content without clear understanding may receive reduced grades or be penalized.

Good Luck 😊